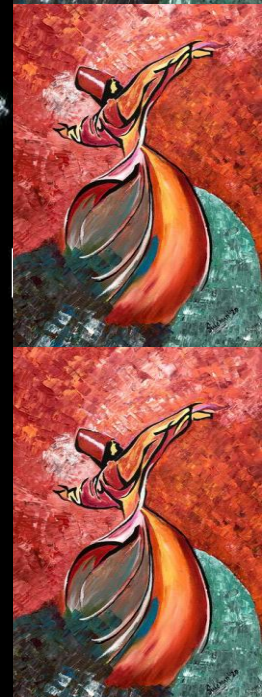




EE 354/CE 361/MATH 310 -  
Introduction to Probability  
and Statistics  
Spring 2026



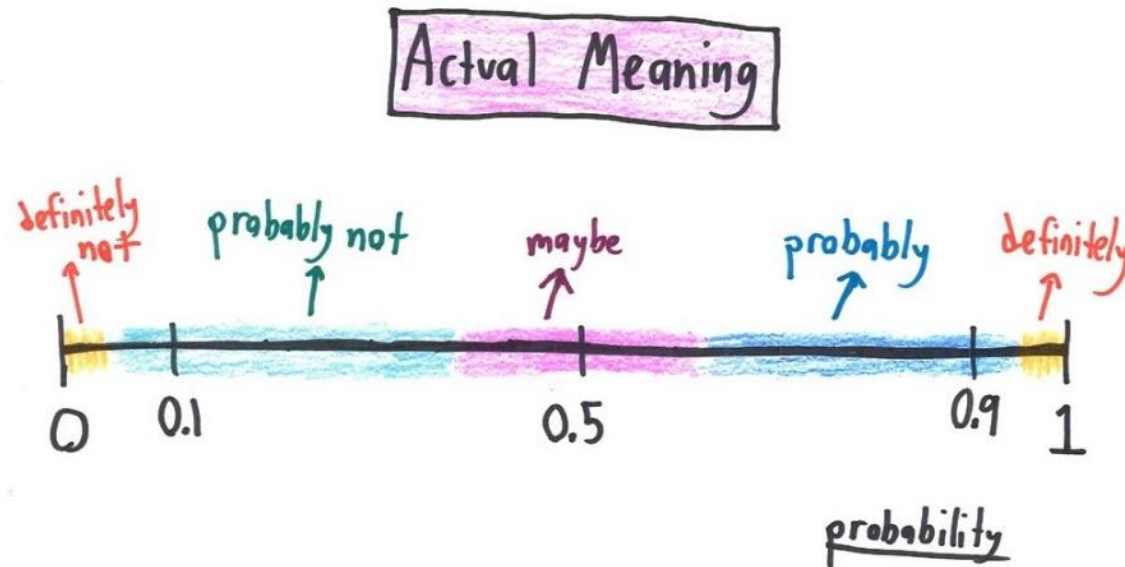


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

## Introduction to Probability and Statistics -

Aamir Hasan

### Week 2 - Lecture No 04 – January 21, 2026



# Announcements!

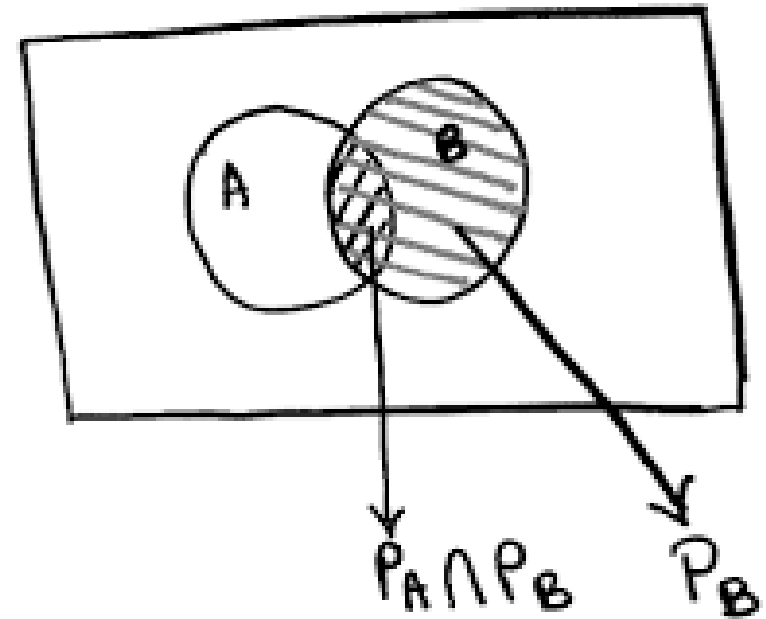
1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. Assignment No 01 to be posted next week

## Agenda for today

1. Quiz No 01 – January 21, 2026
2. Unit 2 – Conditional Probability and Bayes' Rule

## Unit 2: Conditional Probability and Bayes' Rule

- Conditional Probability
- Three important Tools
  - Multiplication rule
  - Total probability theorem
  - Bayes' rule ( $\rightarrow$  Inference)



# Conditioning & Bayes' rule

- Let's say that you roll two dice and their sum comes out to be  $S$ .
  - ❖ If  $S = 9$ , what can we say about the first and second roll?
    - ✓ What is the probability that first roll is 1 or 2?

New knowledge should be used to modify our original beliefs.



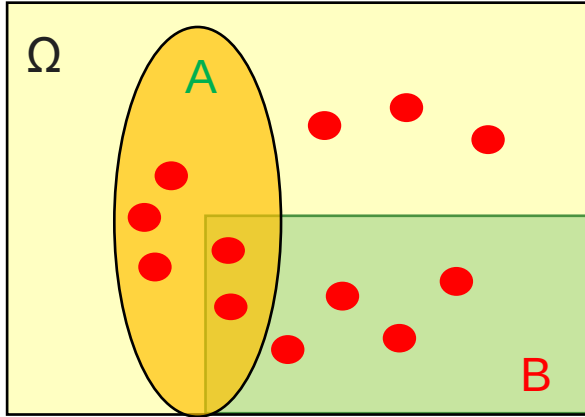
Revised probabilities are called the conditional probabilities.



# The idea of conditioning

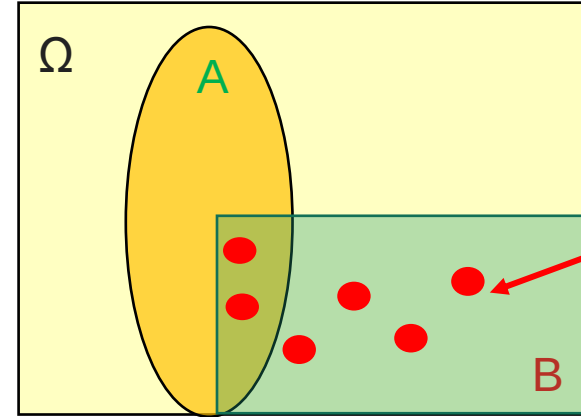
Use new information to revise a model

Assume 12 equally likely outcomes, each with probability  $1/12$



$$P(A) = \frac{5}{12} \quad P(B) = \frac{6}{12}$$

If we are told that  $B$  occurred



$$\frac{1}{6}$$

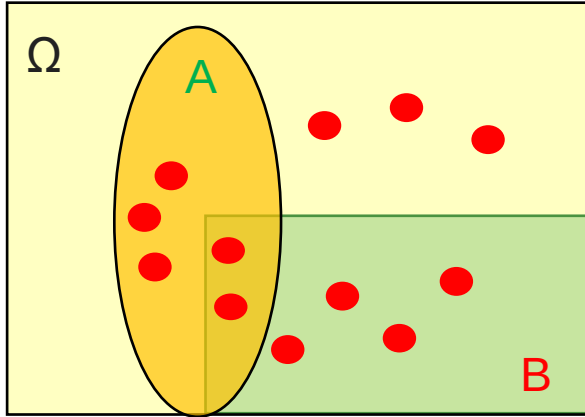
$$P(A|B) = \frac{2}{6} = \frac{1}{3} \quad P(B|B) = \frac{6}{6} = 1$$



# The idea of conditioning

Use new information to revise a model

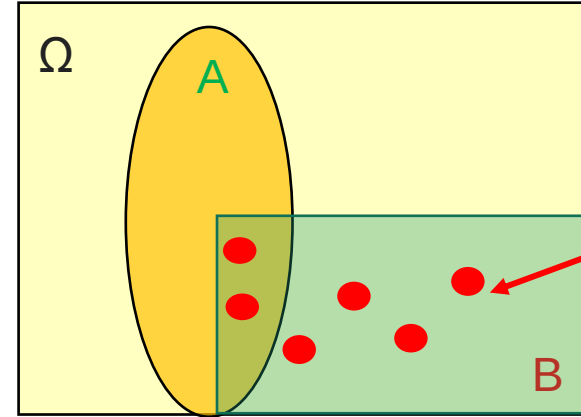
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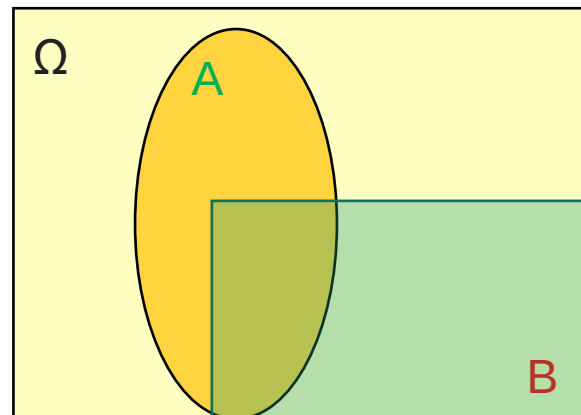
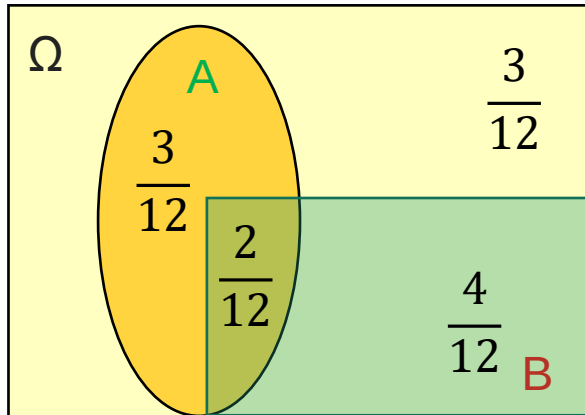
If we are told that  $B$  occurred



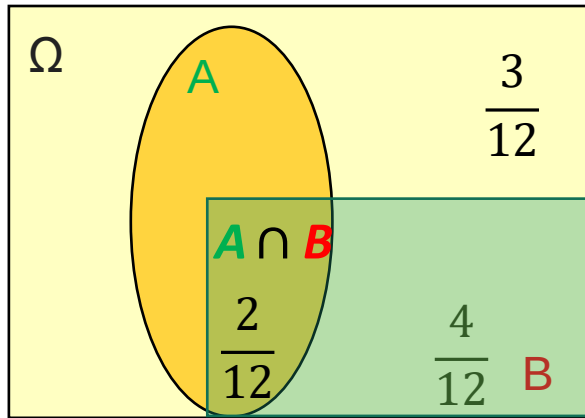
$$\frac{1}{6}$$

$$P(A|B) = \frac{2}{6} = \frac{1}{3} \quad P(B|B) = \frac{6}{6} = 1$$

Assume now that the events are not equally likely



# Definition of conditional probability



- $P(A|B)$  = “probability of  $A$ , given that  $B$  occurred”

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/12}{4/12} = \frac{1}{2}$$

defined only when  $P(B) > 0$



## Example: two rolls of a 4-sided die

Y =  
Second roll

|   |   |   |   |   |
|---|---|---|---|---|
| 4 |   |   |   |   |
| 3 |   |   |   |   |
| 2 |   |   |   |   |
| 1 |   |   |   |   |
|   | 1 | 2 | 3 | 4 |

X = First roll

- Let every possible outcome have probability  $1/16$

- Let  $B$  be the event:  $\min(X, Y) = 2$

Let  $M = \max(X, Y)$

$$\rightarrow P(M = 1 | B) = 0$$

$$\rightarrow P(M = 3 | B) = \frac{P(M = 3 \text{ and } B)}{P(B)}$$

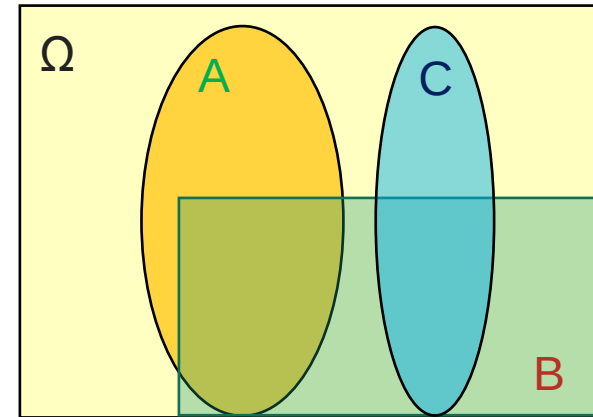
$$\rightarrow = \frac{2/16}{5/16} = \frac{2}{5}$$

# Conditional Probabilities have the same properties as ordinary probabilities

→ •  $P(A|B) \geq 0$  assuming  $P(B) > 0$

→ •  $P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$

→ •  $P(B|B) = \frac{P(B \cap B)}{P(B)} = 1$



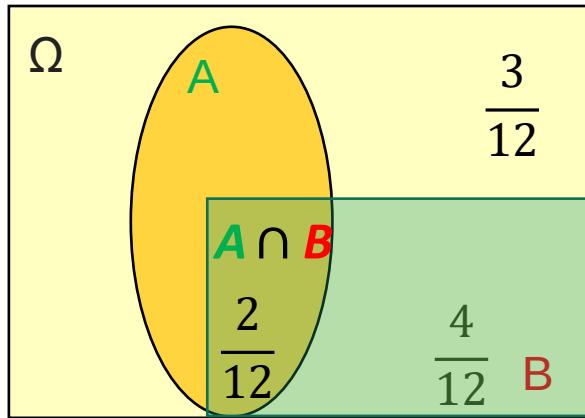
• If  $A \cap C = \emptyset$ , then  $P(A \cup C|B) = P(A|B) + P(C|B)$

→ 
$$= \frac{P((A \cup C) \cap B)}{P(B)} = \frac{P((A \cap B) \cup (C \cap B))}{P(B)} = \frac{P(A \cap B) + P(C \cap B)}{P(B)}$$

→ 
$$= P(A|B) + P(C|B)$$

→ Also finite countable additivity

# Definition of conditional probability



- $P(A|B)$  = “probability of  $A$ , given that  $B$  occurred”

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/12}{4/12} = \frac{1}{2}$$

defined only when  $P(B) > 0$

## Example: Models based on conditional probabilities.

- Event **A**: Airplane is flying above.
- Event **B**: Something registers on radar screen

❖ Problem - Find The probability an aircraft is flying given radar registers something on radar screen –  $P(A|B)$  ?

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\rightarrow \bullet P(A \cap B) = P(A) * P(B|A) = 0.05 * 0.99$$

$$\rightarrow \bullet P(B) = 0.05 * 0.99 + 0.95 * 0.1 = 0.1445$$

$$\rightarrow \bullet P(A|B) = \frac{0.05 * 0.99}{0.1445} = 0.34$$

